



CC3003-1 – WIRELESS NETWORKS AND MOBILE COMPUTING
(For CCE)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

20 Marks

- 1-

10. Which of the following is NOT a requirement for a termination detection algorithm?
 A) All processes must have a known state B) Processes must communicate with each other
 C) There must be a global clock in the system D) The system must be able to detect idle processes
11. Which of the following algorithms is used to detect global termination in mobile distributed systems?
 A) Huang's algorithm B) Snapshot algorithm
 C) Rabin-Karp algorithm D) Ford-Fulkerson algorithm
12. What is the primary role of data synchronization in mobile databases?
 A) To prevent database fragmentation B) To enhance data security in mobile devices
 C) To increase the speed of data retrieval D) To maintain consistency between the mobile device and central database
13. Which of the following best describes the push data delivery model in mobile communication?
 A) Data is requested by the user on demand B) Data is automatically sent to the user without a request
 C) Data is sent only during peak hours D) Data is broadcast to all devices simultaneously.
14. Which of the following scenarios would benefit most from a push data delivery model?
 A) A mobile user streaming a video B) A user browsing for content online
 C) Downloading software updates manually D) A weather application delivering severe weather alerts
15. In the broadcast data delivery model, how is data delivered to mobile users?
 A) Data is delivered to all users within a specific geographic region B) Data is stored temporarily on the server
 C) Data is sent to users based on individual subscriptions D) Data is sent to a specific user after a request
16. Which of the following best describes on-demand data delivery in mobile communication?
 A) Data is pre-loaded on the device and accessed later B) Data is sent continuously to the user in real time
 C) Data is sent to a group of users simultaneously D) Data is delivered to users only when explicitly requested
17. Cloud computing is a kind of abstraction which is based on the notion of combining physical resources and represents them as _____ resources to users.
 A) Real B) Virtual
 C) Cloud D) All mentioned
18. In mobile databases, which technique is commonly used to reduce the impact of network disconnections?
 A) Data normalization B) Encryption
 C) Data fragmentation D) Data replication
19. What is the main challenge faced by mobile transactions compared to traditional transactions?
 A) High cost of mobile storage B) Lack of support for SQL queries
 C) Limited computing power of mobile devices D) Network unreliability and frequent disconnections
20. In mobile transaction processing, what is the role of the commit point?
 A) To mark the point at which a transaction is completed successfully B) To handle transaction rollback
 C) To check if the database is synchronized D) To validate the transaction's correctness

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Differentiate between the eMBB, mMTC, and URLLC for the 5G performance requirement.	8	L*2	1	1
	b) Elaborate the importance of Mobile Computing.	8	L2	2	1
2.	a) Explain the features for 4G mobile communication system.	8	L2	1	1

- b) A mobile cellular network is designed with the following parameters:
 Average number of calls in a cell: 150
 Average call duration: 120 seconds
 Average channel holding time: 90 seconds
 Arrival rate of originating calls: 0.8 calls/sec
 Probability of a new call requiring handoff: 0.9
 Probability of a handoff call requiring another handoff: 0.85
 Blocking probability of originating calls: 0.1
 calculate the following: The arrival rate of handoff calls, the outgoing rate of mobile users and the probability of handoff failure.

8 L3 2 1

3. a) Discuss the anatomy of Next generation Smart phones.
 b) Discuss the Heuristics technique for solving fixed channel assignments problems.

8 L2 1 1

8 L2 2 1

Unit – II

4. a) Explain the push and pull based mechanism with block diagram.
 b) The given 10 data items, with their popularity ratios decreasing from D1 to D10. The partition into three disks with the following assignments and relative frequencies:
 Disk 1: Data items D1, D2 (Relative frequency = 4)
 Disk 2: Data items D3, D4, D5, D6 (Relative frequency = 2)
 Disk 3: Data items D7, D8, D9, D10 (Relative frequency = 1)
 Determine the max_chunks, calculate the number of chunks for each disk and derive the broadcast schedule for one full cycle.

8 L2 3 1

8 L3 3 1

5. a) Discuss the effect of cache in mobile computing.
 b) Termination detection algorithms assume that an idle process can only be activated on the reception of a message. Consider a system where an idle process can become active spontaneously without receiving a message. Do you think a termination detection algorithm can be designed for such a system? Give reasons for your answer.

8 L2 3 1

8 L3 4 1

6. a) A data server has three data items, D₁, D₂, and D₃, with popularity ratios $p_1=0.6$, $p_2=0.3$, and $p_3=0.1$, respectively. To achieve minimum average access latency, the server needs to determine the optimal broadcast frequencies for each item. Calculate the optimal broadcast frequencies for each data item.
 b) Design an efficient termination detection algorithm for a system where the computation at a process is instantaneous.

8 L3 3 1

8 L3 4 1

Unit – III

7. a) With suitable diagram explain the high-level view of process migration.
 b) What is cloud computing. Explain the essential characteristics of it.
 8. a) Elaborate the role of Dynamic host configuration protocol (DHCP) server in mobile computing.
 b) Why the multi-attribute queries are desirable? Explain it.

8 L2 5 1

8 L2 5 1

8 L2 5 1

8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
